Classical and quantum harmonic oscillators

$$H = \frac{p^2}{2m} + \frac{1}{2} m\omega^2 x^2$$

$$Z_{cl} = \frac{1}{2\pi\hbar} \int dp\,dx\; e^{-\frac{p^2}{2mkT}}\, e^{-\frac{m\omega^2 x^2}{2kT}} = \frac{kT}{\hbar\omega}$$

$$E = \hbar\omega(n+\tfrac{1}{2})$$

$$Z_q = \sum_{n=0}^{\infty} \exp\left[-\frac{\hbar\omega}{kT}\left(n+\frac{1}{2}\right)\right] = e^{-\frac{\hbar\omega}{2kT}} \left\{ 1 + e^{-\frac{\hbar\omega}{kT}} + \left(e^{-\frac{\hbar\omega}{kT}}\right)^2 + \dots \right\}$$

$$= \frac{e^{-\frac{\hbar\omega}{2kT}}}{1-e^{-\frac{\hbar\omega}{kT}}} = \frac{1}{e^{\frac{\hbar\omega}{2kT}} - e^{-\frac{\hbar\omega}{2kT}}} = \frac{1}{2\,\mathrm{sh}\left(\frac{\hbar\omega}{2kT}\right)}$$

Classical limit $\hbar \to 0 \Rightarrow Z_q \to \frac{kT}{\hbar\omega}$
$(\hbar\omega \ll kT)$

Quantum limit $\hbar\omega \gg kT$: $Z_q \to e^{-\frac{\hbar\omega}{2kT}}$

N oscillators: $Z_q = \left(\frac{1}{2\,\mathrm{sh}\frac{\hbar\omega}{2kT}}\right)^N$

$$Z = e^{-F/kT} \Rightarrow F = -kT \ln Z$$

$$dF = -S\,dT - p\,dV \Rightarrow S = -\left(\frac{\partial F}{\partial T}\right)_V$$

$$F = +kT N \ln\left(2\,\mathrm{sh}\frac{\hbar\omega}{kT}\right)$$

$$U = \frac{\sum_{n=0}^{\infty} E_n e^{-E_n/kT}}{Z_q} = \left(\beta \equiv \frac{1}{kT}\right) = \frac{\sum -\frac{\partial}{\partial\beta} e^{-E_n\beta}}{Z}$$

$$= -\frac{\partial}{\partial\beta} \ln Z = -\frac{\partial Z}{\partial\beta}\frac{1}{Z}.$$

$$U = -\frac{\partial}{\partial\beta} \ln \frac{1}{2^N \operatorname{sh}^N \beta\hbar\omega} = +N \frac{\partial}{\partial\beta} \ln\left(2 \operatorname{sh} \beta\frac{\hbar\omega}{2}\right) =$$

$$= \frac{N \operatorname{ch}\left(\beta\frac{\hbar\omega}{2}\right)}{\operatorname{sh}\left(\beta\frac{\hbar\omega}{2}\right)} \cdot \frac{\hbar\omega}{2} = N \frac{\hbar\omega}{2} \frac{e^{\beta\frac{\hbar\omega}{2}} + e^{-\beta\frac{\hbar\omega}{2}}}{e^{\beta\frac{\hbar\omega}{2}} - e^{-\beta\frac{\hbar\omega}{2}}} =$$

$$= N\frac{\hbar\omega}{2} \frac{e^{\beta\hbar\omega}+1}{e^{\beta\hbar\omega}-1} = N\frac{\hbar\omega}{2} \frac{e^{\beta\hbar\omega}-1+2}{e^{\beta\hbar\omega}-1} =$$

$$= N\frac{\hbar\omega}{2}\left(1 + \frac{2}{e^{\beta\hbar\omega}-1}\right) = N\left(\frac{\hbar\omega}{2} + \frac{\hbar\omega}{e^{\beta\hbar\omega}-1}\right)$$

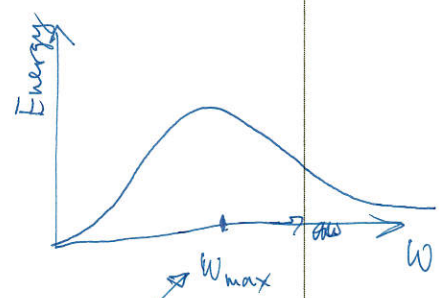
$\Rightarrow$ total internal energy of 1d gas of quantum oscillators

$$U \sim \int d\omega \; \underbrace{\hbar\omega}_{\text{energy}} \cdot \underbrace{\frac{1}{e^{\frac{\hbar\omega}{kT}}-1}}_{\text{number of oscillators}} \sim T^2$$

$\Rightarrow$ for 3d gas $\quad U \sim \int \underbrace{d\omega \cdot \omega^2}_{\text{volume of spherical shell of radius } \omega} \quad \frac{\hbar\omega}{e^{\frac{\hbar\omega}{kT}}-1} \sim T^4$

Planck's distribution $\quad \dfrac{\omega^2 \hbar\omega}{e^{\frac{\hbar\omega}{kT}}-1}$

Cosmic Microwave Background $(T \sim 3K)$



function of T only.